Unified dynamics of training, learning, inference, and reinforcement in neural networks

Mian Wang

Wuhan Polytechnic University, Wuhan, China

Corresponding author. Email: wangmian945@gmail.com

Although training, learning, inference, and reinforcement have been extensively studied, their distinct dynamical roles and connections have lacked a unified experimental characterization. Nine nanoGPT experiments established training–learning–inference relations: realized updates elicit state-dependent finite-amplitude responses; learning persistently reorganizes distributed functional support, with target-specific readout-boundary crossings determining capability transitions; inference recruits learned support without changing the learned state. Factors identified in the fifth experiment yielded a direct predictor of post-update capability transitions, achieving 91.43% accuracy and 91.49% four-way macro recall across 5,088 targets. We tested these relations and prediction across five additional neural architectures. In a seventh system, concentrated correct feedback eroded previously mastered skills; rebalanced feedback restored their accuracy from 70.31% to 99.48%. Linking inference consequences to subsequent training closes these relations into unified training–learning–inference–reinforcement dynamics.

Training, generalization, capability formation, and inference are central problems in neural-network research. Different fields have developed influential theories to study them.

In *optimization theory*, the Neural Tangent Kernel (NTK) describes parameter-induced output changes through the network Jacobian and the corresponding kernel in specified training regimes (*1*). Edge of Stability studies training behavior in relation to the largest Hessian eigenvalue, learning rate, and loss trajectory (*2*). Sharpness-Aware Minimization (SAM) constructs training updates

using loss changes in a parameter neighborhood to improve generalization (*3*). Here, the *current state* is represented by parameters and their local derivatives or curvature; *realized updates* are produced by the optimizer; and *responses* appear as changes in model outputs or loss.

In *representation learning and generalization*, Spectral Bias examines frequency-dependent learning rates of functions or kernel modes (*4*). Implicit Bias studies why an optimizer selects particular solutions, such as maximum-margin directions on separable data (*5*). Neural Collapse measures class-feature means, within-class variability, and classifier geometry, revealing continuing representational change after training accuracy saturates (*6*). Information Bottleneck theory studies mutual information among inputs, hidden representations, and targets, including the retention and compression of task-relevant information (*7*). Their *internal organizations* take the form of spectral modes, optimizer-selected solutions, feature geometry, or information structure; their *functional changes* appear in learning rates, evolving solutions, and predictive performance.

In *capability formation*, Grokking captures the long delay between fitting training examples and achieving strong test generalization (*8*). Subsequent circuit studies track the formation and reorganization of internal computations during this delay (*9*). Double Descent studies how test error can rise near interpolation and fall again as model size, training time, or data conditions change (*10*, *11*). These studies measure the *current state* through training phase or interpolation regime, *internal change* through circuit and solution evolution, and *observable capability* through accuracy and generalization error.

In *continual learning*, catastrophic forgetting occurs when training on a new task reduces performance on earlier tasks. Elastic Weight Consolidation (EWC) estimates parameter importance using Fisher information from earlier tasks and penalizes changes to important parameters (*12*). Gradient Episodic Memory (GEM) constrains new-task gradients using their inner products with previous-task gradients (*13*). Orthogonal Gradient Descent (OGD) projects new gradients into directions that approximately preserve earlier-task outputs (*14*). In these methods, the *existing state* includes previously learned parameters and function-sensitive directions; *updates* are constrained new-task actions; *interactions* occur through overlapping gradient or functional subspaces; and *capability changes* are measured as retention, forgetting, or transfer.

In *inference research*, Attention uses queries, keys, and values to compute input-dependent information weights (*15*). In-context learning studies how examples and task information provided

in the context affect predictions while model parameters remain fixed (*16*). Chain-of-Thought prompting studies how intermediate reasoning steps affect performance on complex tasks (*17*). The *current state* is the trained model; *input conditions* include the query, context, and intermediate computations; *internal computation* varies with attention and activations; and *functional performance* is observed in the resulting predictions.

In *reinforcement and feedback-driven training*, reinforcement learning from human feedback (RLHF) uses preference and reward signals to adjust model behavior (*18*). Self-training creates further training data from a model's predictions (*19*). Recursive model-collapse studies examine the loss of low-probability modes when successive models are trained on data generated by their predecessors (*20*). Here, *feedback* is supplied through rewards, preferences, or generated-data distributions; *training actions* are the resulting updates; and subsequent *functional change* is observed through behavior, accuracy, and generated distributions.

These studies use different mathematical objects—kernels, Hessians, gradients, class features, mutual information, task scores, attention weights, and reward signals—yet six recurring categories can be identified: *current state, realized update, functional response, internal organization, observable capability, and feedback. The core TLIR relations were established experimentally before this cross-field comparison; identifying these recurring variables here is a way to help readers understand the theory, not the process by which it was discovered.*

Figure 1 illustrates the four linked dynamical roles.

Experimental Establishment of TLI Open-Loop Dynamics

The nine nanoGPT experiments that established TLI involved a substantially broader set of native interventions, controlled comparisons, and independent validations than can be presented within the main text. This section summarizes their key experimental findings and the resulting theoretical relations. Complete protocols, execution records, negative results, and independent validators are provided in the accompanying repository.

Experimental Instruments and Evidence Control

These experiments required instruments that were developed as the measurement questions arose, rather than assumed before the theory. Their common evidence substrate was the Generation-Fact Graph (GFG), which links recorded model and optimizer states, actual update occurrences, interventions, outputs, and their source identities. For Training, realized-update causal forks restored matched receiving states and compared full, skipped, parameter-only, optimizer-memory, and reciprocal transfers, separating an update from the state on which it acts. A finite-amplitude path instrument then applied the same recorded update at frozen amplitude values to measure nonlinear numerical responses and discrete transitions. For Learning, CSRG-4C used repeated baseline, single-component, and paired-component gates to measure necessity, backup, substitution, synergy, and failure slack in the functional support of registered target groups. Finally, an identity-aligned boundary ledger joined each registered target to the exact pre-update state and realized update, then separated pre-response records from post-update adjudication of correctness. The instruments enabled the subsequent experimental tests; frozen protocols, negative controls, and independent validators constrained their interpretation.

Training: Realized Updates and Receiving States

At a registered training step, the parameter–optimizer state is $Z_t = (\theta_t, \omega_t)$, and the realized parameter update is $\Delta\theta_t = \theta_{t+1} - \theta_t$.

TL-E01 revealed capability formation, decline, and recovery that could not be fully characterized by loss, accuracy, or training step alone. TL-E02 further showed that pausing effective parameter–optimizer evolution delays capability formation, while the clipping-threshold control did not reproduce this effect.

TL-E03 used full/skip branches and reciprocal update transplants to establish that the same realized update produces different functional responses in different receiving states:

$$R_q(\theta_A, \Delta\theta_B) \quad \text{vs.} \quad R_q(\theta_B, \Delta\theta_B), \tag{1}$$

where $R_q(\theta, \Delta) = m_q(\theta + \Delta) - m_q(\theta)$. Separate optimizer-memory exchanges examined how his-

torical optimizer states affect update formation:

$$\Delta^{(a,b)} = \mathrm{OptStep}(\theta_t, g_t, \omega_{a,b}). \tag{2}$$

TL-E04 measured the complete finite-amplitude response to an actual update:

$$r_{t,q}(\alpha) = m_q(\theta_t + \alpha\Delta\theta_t) - m_q(\theta_t). \tag{3}$$

The experiments identified near-linear, saturating, accelerating, turnback, and sign-reversal responses. TL-E05 subsequently identified three primary response-conditioning families: target-boundary state (F1), target-specific update geometry (F3), and parameter–optimizer receiving state (F5).

Learning: Distributed Support and Capability Boundaries

TL-E06 used reversible component interventions to establish distributed functional support, including component necessity, backup, substitution, synergy, and failure slack.

TL-E07 then examined how realized training updates persistently reorganize this support:

$$\Delta P_{t,q} = P_q(\theta_t + \Delta\theta_t) - P_q(\theta_t). \tag{4}$$

Across 72 experimental sections, 1,636 valid support-allocation transitions and 228 primary-support hand-offs were recorded.

TL-E08 linked these internal changes to observable capability through identity-aligned target readout boundaries. The signs of the pre- and post-update margins distinguish maintenance, loss, non-acquisition, and acquisition or recovery. Learning is thus identified with persistent functional reorganization, while capability transitions are determined by crossings of target-specific readout boundaries.

Inference: Frozen Recruitment of Learned Support

INF-E01 examined 13 nanoGPT training histories and 52 checkpoints. With persistent learned states frozen, the experiment measured query-conditioned component effects, non-additive support interactions, and changes produced by replacing trained components with their exact pre-formation versions.

At a fixed learned state L^*, inference recruits and combines learned support:

$$S_{q,c}^{\text{active}} = \Pi_{q,c}(L^*), \qquad y_{q,c} = H_{q,c}(S_{q,c}^{\text{active}}). \tag{5}$$

Non-additive component interactions are measured by

$$I_{ij,q} = f_{\text{full}} - f_{-i} - f_{-j} + f_{-i,-j}, \tag{6}$$

while learned-version dependence is tested through

$$D_{j,q} = f(q;\theta^*) - f(q;\theta^{(j\leftarrow\text{pre})}). \tag{7}$$

Exact restoration verified that these interventions did not persistently alter the learned state. The experiments establish Inference as the query-conditioned recruitment and combination of support formed through Learning.

Together, these nine experiments established the connected Training–Learning–Inference relations, forming the TLI open-loop dynamics. Figure 2 summarizes the experimental evidence across all four TLIR roles.

Prediction and Cross-System Validation

Theory-Derived Target-Boundary Prediction

The response-factor analysis in TL-E05 identified three primary conditioning factors: the current target-boundary state (F1), target-specific update geometry (F3), and parameter–optimizer receiving state (F5). These findings led directly to TL-P01, a predictor of immediate target-boundary outcomes following an actual formed update.

For target y, competitor c, and hidden representation h, let $b_c = w_y - w_c$ and $\Delta b_c = \Delta w_y - \Delta w_c$. Using the first and second directional derivatives $h' = Dh[\Delta\theta_t]$ and $h'' = D^2h[\Delta\theta_t, \Delta\theta_t]$, the predictor computes

$$\begin{aligned}\widehat{g}_{y,c} = b_c^\top h + \Delta b_c^\top h + b_c^\top h' \\ + \Delta b_c^\top h' + \tfrac{1}{2} b_c^\top h''.\end{aligned} \tag{8}$$

The target is predicted correct if $\min_{c\neq y} \widehat{g}_{y,c} > 0$. Combining predicted post-update correctness with the known pre-update state gives four capability transitions. Predictions use the already formed update and pre-response information, without reading actual post-update target outputs.

Cross-System Validation

We subsequently tested the established TLI relations and their predictive consequences in five additional neural systems. Table 1 summarizes the seven systems used in the study, including the separate GRU reinforcement experiment examined in the reinforcement section.

Each of the five additional TLI systems was examined through realized-update interventions, receiving-state comparisons, finite-amplitude responses, component-coalition support measurements, target-boundary outcomes, and frozen-inference tests. Cross-system implementations evaluated all 16 coalitions of four registered components, with exact Shapley effects:

$$\phi_{j,q} = \sum_{C \subseteq \mathscr{C} \setminus \{j\}} \frac{|C|!(3-|C|)!}{4!} \left[v_q(C \cup \{j\}) - v_q(C) \right]. \tag{9}$$

Here $v_q(C)$ denotes the target margin under active component coalition C.

All 42 ResNet, 18 diffusion U-Net, and 36 Vision-MoE parameter receiving-state exchanges exceeded their registered response-difference thresholds. Finite-amplitude nonlinear responses occurred in 32.23%, 19.25%, and 35.71% of their respective target-update records, while support reallocation exceeded its specified threshold in 91.41%, 97.02%, and 80.43%. Their frozen-inference experiments further confirmed query-conditioned support, non-additive interactions, and dependence on exact trained component versions.

The GNN reproduced the main TLI relations across three seeds, although one seed failed the registered nonlinear-response frequency criterion (1.56%). The hard-spiking SNN passed all twelve registered TLI criteria across four seeds, extending the evidence to discontinuous spiking dynamics.

The prediction algorithm was also evaluated across these systems, with task-specific adaptations of the directional response computation. Table 2 reports immediate post-update target-boundary classification.

The predictor achieved high four-way recall in five systems. In the hard-spiking SNN, discontinuous activation thresholds limited local derivative-based prediction, yielding 70.74% macro recall despite 95.31% accuracy. Together, the experiments demonstrate cross-system recurrence of the core TLI relations and identify architecture-dependent differences in their quantitative responses and predictive approximations. Figure 3 summarizes the six within-system TLI experiment chains.

Reinforcement: From TLI Open-Loop to TLIR Closed-Loop Dynamics

Theory-Derived Prediction of Feedback Dynamics

The established TLI relations show that training persistently reorganizes distributed functional support, while inference recruits and combines this learned support to produce behavior. When the consequences of inference are repeatedly fed back into subsequent training, the resulting updates can further reorganize the support on which future inference depends.

This led to a prediction: sustained, narrowly concentrated correct positive feedback could progressively favor the reinforced capability while weakening other already mastered capabilities. The predicted effect arises from feedback-directed functional reorganization, rather than incorrect rewards.

Experimental Tests and Results

RL-E05/E06 tested this prediction in a seventh system, a shared attention-free GRU policy. All four skills reached mastery before the experimental branches began. Every branch started from an identical parameter–AdamW state, and each complete two-decision episode received $r = 1$ only when correct, and $r = 0$ otherwise.

RL-E05 compared selective, balanced, and frozen feedback. RL-E06 further varied the allocation of 60 episodes per update over 3,200 updates (Table 3).

RL-E05 observed relative support concentration, cross-skill interference, and component-version-dependent trade-offs. Its stronger preregistered temporal-precedence criterion was not met.

In RL-E06, all 12 formal seeds exhibited dose-ordered changes. The reinforced skill's mean support share increased from 25.54% under balanced feedback to 35.20% under exclusive feedback, while the other skills' mean margins decreased from 9.474 to 3.962. Mild and high concentration already reduced these margins without reducing accuracy. Exclusive feedback decreased other-skill accuracy by 39.06 percentage points relative to balanced feedback.

Two recovery branches then started from the exact exclusive-feedback state at update 800.

Rebalanced feedback restored other-skill accuracy from 70.31% to 99.48% while preserving the reinforced skill at 100%. Training only the other three skills restored their accuracy to 99.83%, but reduced the previously reinforced skill to 89.06%.

These experiments establish that, in the tested shared system, the distribution of correct feedback can redirect persistent functional support, alter previously mastered capabilities, and reverse these changes through feedback redistribution.

TLIR Closed-Loop Dynamics

Reinforcement connects the consequences of inference to subsequent training actions. The resulting training updates reorganize learned support, which in turn changes future inference and its consequences. This feedback closure transforms the experimentally established TLI open-loop dynamics into the unified Training–Learning–Inference–Reinforcement (TLIR) closed-loop dynamics.

Unified Explanations of Positive and Negative Findings

Independent historical stress tests. Only after the experimental establishment of TLI and its reinforcement closure did we compare TLIR with 21 previously published positive/negative pairs. These works were not the design set for TLIR. The question is whether one receiver–action–response–support–boundary–feedback relation can explain both why a local theory succeeds and why its extrapolation fails. Each pairing requires a specific changed mechanism, an internal experimental anchor, and a possible falsifier; merely invoking *different conditions* is insufficient. These comparisons provide external explanatory convergence, not new causal verification of TLIR by the original authors.

Training: kernel dynamics and feature formation. The neural tangent kernel gives a rigorous account in specified infinite-width lazy-training regimes (*1*), whereas feature-learning parameterizations admit nontrivial representation evolution (*21*). Both can hold because parameterization and receiving state govern the functional response to an actual update. TL-E03's reciprocal update transplants and TL-E04's finite-amplitude paths anchor this distinction experimentally. A locally exact kernel description does not establish a universal closure for finite updates and evolving support.

A robust finite-kernel closure across different receivers and actual actions would instead challenge this interpretation.

Learning: delayed capability and continuing formation. Grokking exhibits late test generalization after training performance has saturated (*8*); analyses of its circuits reveal formation before the observed transition (*9*). Neural collapse likewise documents continuing feature organization after zero training error (*6*). TL-E07/E08 separate persistent reorganization from a target-specific readout: internal support and margins may evolve without changing a discrete accuracy measure until a boundary is crossed. Learning therefore neither begins nor ends at a single accuracy threshold.

Inference: attention's causal role versus attention-weight explanations. Attention implements query-conditioned information combination (*15*). Yet attention weights need not agree with feature-importance measures and can be altered without equivalent output changes (*22*); other controlled tests show that these observations do not make attention universally uninterpretable (*23*). These results concern different explanatory objects. INF-E01's exact-version rollback, component gates, and non-additive pair effects establish a causal dependence of frozen inference on recruited learned support, without equating a single attention-weight map with that entire support. The intervention-defined support relation, rather than a particular attribution score, is the object being tested.

Reinforcement: correct feedback and cross-capability trade-offs. Preference-based feedback can improve target behavior (*18*), but success on a reinforced objective need not preserve all other functions of a shared system. RL-E06 isolated this distinction using *correct* task rewards, rather than a mis-specified reward proxy: increasing feedback concentration systematically redirected support and reduced other-skill margins, while balanced feedback reversed much of the loss. RL-E05's stronger preregistered temporal-precedence criterion was not met; its failure is distinct from the subsequently confirmed dose–duration–recovery relations. Thus a locally beneficial feedback process can be globally double-edged without implying that every reinforcement learner behaves identically. Other stress-test pairs locate sharpness/SAM (*3*, *24*) and information-bottleneck compression (*25*, *26*) as potentially useful local mechanisms rather than necessary descriptions of all learning.

Theory-first conditional extrapolations. Separate from those historical comparisons, we derived conditional consequences of the experimentally grounded TLIR relations before consulting the corresponding literature. This derivation-before-search procedure is not equivalent to blinded prospective preregistration. Agreement with independent results therefore supports the scope of the

deductions, not a claim that the cited papers tested TLIR itself.

Evolving stability. For parameter-only gradient descent on a locally quadratic loss, a positive-curvature frozen Hessian mode obeys the step factor $1-\eta\lambda_i$, giving the local stability condition $|1-\eta\lambda_i|<1$ and upper boundary $\lambda_i = 2/\eta$. TLIR places that condition inside a changing receiver–action map: finite updates and optimizer memory can change subsequent curvature and responses, so exceeding a frozen local boundary does not by itself establish divergent long-horizon learning. Observed edge-of-stability trajectories provide an external comparison (*2*).

Post-capability learning. In separable logistic classification, all target margins $m_i = y_i w^\top x_i$ may be positive while gradients remain nonzero. For large margins, the residual training action weights examples approximately by e^{-m_i}, emphasizing those closest to the boundary. Under the corresponding linear-separability and loss-tail assumptions, continued training selects the maximum-margin direction (*5*). This is a conditional deduction from the distinction between observable correctness and persistent learning, not a claim that every neural architecture converges to the same margin law.

Recursive feedback. If successive generated training sets retain a mode with factor $0<\rho<1$ and replace earlier data, its mass contracts. If instead a fraction α of external data with mode mass $q_{\text{real}}>0$ persists, the conditional recurrence is

$$q_{t+1} = (1-\alpha)\rho q_t + \alpha q_{\text{real}}. \tag{10}$$

For $\alpha = 0$ the mode decays; for $\alpha > 0$ it approaches $q^* = \alpha q_{\text{real}}/[1-(1-\alpha)\rho] > 0$. Beneficial self-training, recursive tail loss, and protection from continuous real-data inclusion (*19*, *20*, *27*) can thus occupy different feedback regimes rather than contradict each other.

These derivations state assumptions and boundaries rather than universal numerical laws. We invite reviewers and independent researchers to derive additional conditional predictions from TLIR, specify in advance what outcomes would falsify them, and test those predictions in new systems.

Conclusion

Through systematic experiments across neural systems, this work reveals dynamical relations spanning neural-network training, learning, inference, and reinforcement. From capability formation

and use to loss and recovery, TLIR establishes experimentally testable causal connections between observable performance changes and the organization and evolution of internal functional support. Neural-network capability evolution thus moves beyond a black-box performance phenomenon and becomes a dynamical process that can be investigated through precise experimental interventions and quantitative prediction.

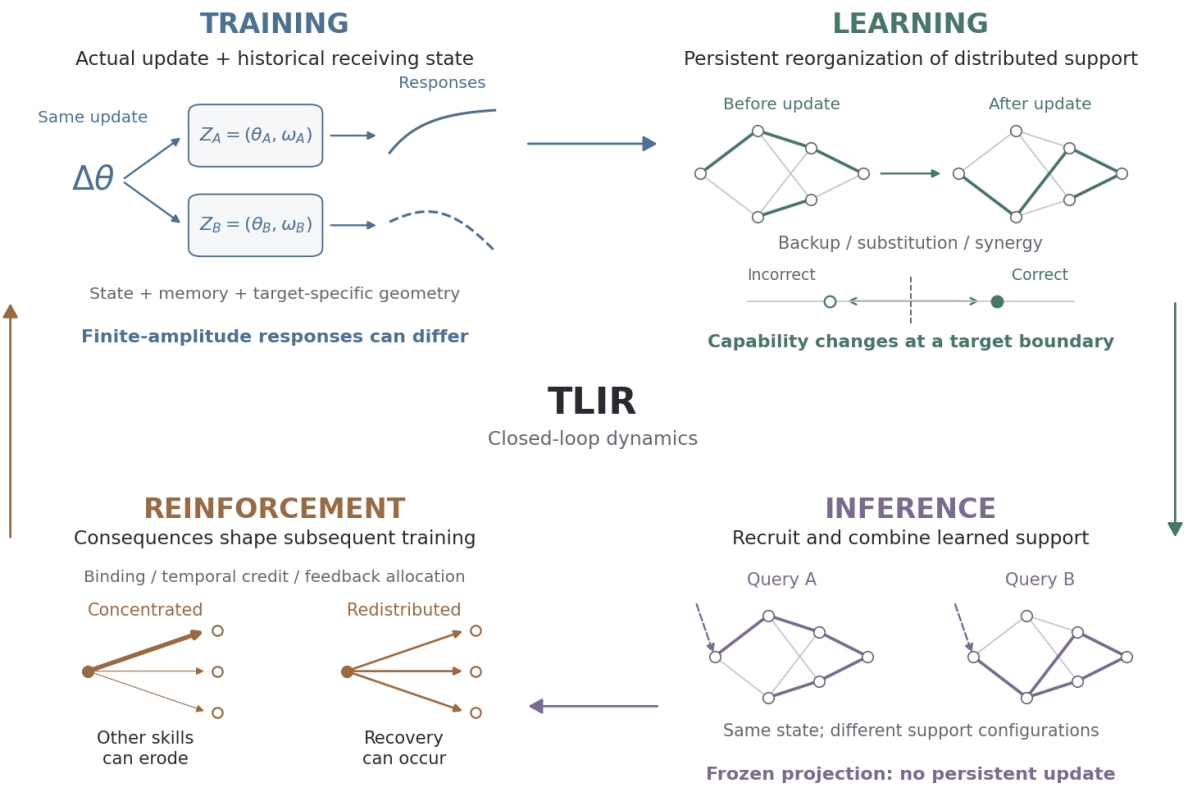


Fig. 1. TLIR closed-loop dynamics. Training, Learning, and Inference establish the open-loop core; Reinforcement feeds behavioral consequences into subsequent training.

Table 1. Seven neural systems: six TLI chains and a GRU feedback study.

System (optimizer)	Task / registered components
nanoGPT (Adam)	Cyclic next-token; 4 attention/MLP branches
ResNet-18 (SGD momentum)	CIFAR-100 classification; 4 residual stages
Diffusion U-Net (AdamW)	CIFAR-10 denoising; 4 U-Net paths
Vision-MoE (AdamW)	CIFAR-100 classification; 4 routed experts
GNN (RMSprop)	Graph reachability; 4 message-passing blocks
Spiking SNN (Adagrad)	8×8 digits; 4 spiking blocks
GRU (RL-E05/E06; AdamW)	Four-skill feedback; 4 shared hidden groups



Fig. 2. Selected evidence for Training, Learning, and frozen Inference in nanoGPT, and feedback-driven capability change and recovery in the shared GRU. (a1) Paused evolution delays capability formation; (a2) identical updates elicit different receiver responses; (a3) finite-amplitude response types; (a4) boundary, update, and receiver factors. (b1) Joint component removal crosses a target boundary; (b2) a realized update reallocates support; (b3) pre- and post-update target margins. (c1) Query-conditioned component effects; (c2) non-additive pair effects; (c3) learned-version rollback and exact restoration. (d1) Feedback-dose support redistribution; (d2) margin erosion before accuracy loss; (d3) rollback trade-offs; (d4) feedback rebalancing and recovery.

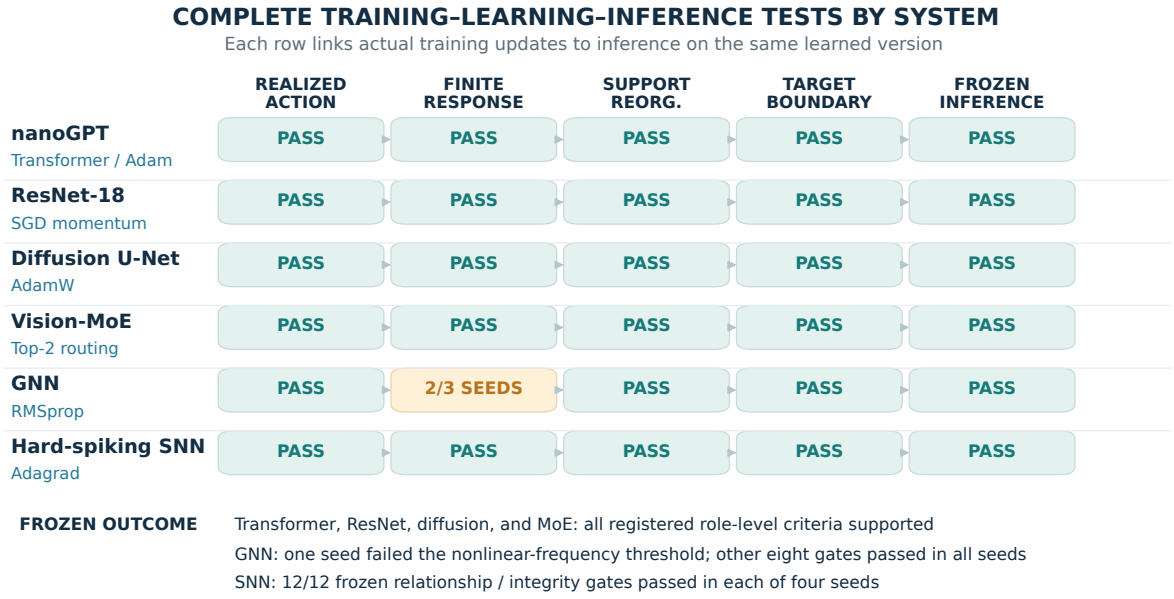


Fig. 3. Within-system training–learning–inference experiment chains. One GNN seed meets eight of nine registered gates, with nonlinear-response incidence below its specified threshold; SNN passes all twelve registered gates in four seeds.

Table 2. Post-update target-boundary classification reports. Macro recall averages four observed transition categories; high accuracy alone is not sufficient. Predictor details and evaluation sets differ by system.

System	Accuracy (%)	Four-way recall (%)
nanoGPT	91.43	91.49
ResNet-18	99.28	97.99
Diffusion U-Net	99.60	97.98
Vision-MoE	99.63	96.95
GNN[a]	98.54	98.62
Hard-spiking SNN[a]	95.31	70.74

[a]Independent prediction tests use registered GNN/SNN training histories.

Table 3. RL-E06 feedback episodes per update, by capability.

Feedback	q_0	q_1	q_2	q_3
Balanced	15	15	15	15
Mild	30	10	10	10
High	45	5	5	5
Exclusive	60	0	0	0

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Author contributions: Mian Wang conceived the study; developed the atomic generation-fact model, the Generation-Fact Graph, the GFG-based recursive scientific process, and the unified theory of training, learning, inference, and reinforcement; defined the scientific questions, research boundaries, and decision criteria; introduced the principal theoretical hypotheses and conceptual transitions; directed the research program; analyzed and interpreted validated results; developed

prospective predictions and theoretical deductions; prepared the figures; and wrote and revised the manuscript.

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Data, code and materials availability: The study's public evidence repository is organized by the scientific claims of this paper to support rapid machine-assisted auditing (https://github.com/wind342/tlir-research-evidence/tree/codex/tlir-paper-evidence-navigation-v1). A paper-to-evidence map connects the main claims, experiments, figure panels, and reported results to frozen protocols, source code, result records, provenance and checksum manifests, and independent verification procedures. The public evidence matrix specifies which results can be independently recomputed and which underlying checkpoints or runtime artifacts are not included in the public release.

Use of generative artificial intelligence: General-purpose AI systems, including OpenAI Codex and ChatGPT, served as scientific agents within the GFG-based recursive scientific process. Under the author's direction, these AI systems invented and implemented the four experimental instruments—CSRG-4C, realized-update causal forks, finite-amplitude update paths, and the identity-aligned target-boundary ledger—and designed, implemented, initiated, monitored, and executed the reported experiments. They performed analyses, causal interventions, matched replays, counterexample searches, and independent validation procedures, and contributed to manuscript drafting, editing, and LaTeX formatting. The reported numerical results were produced by the documented native programs and execution environments, not generated directly as text by AI models. The author defined the scientific questions, research boundaries, and decision criteria, reviewed the validated evidence, and accepts responsibility for the scientific claims and results.